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RESEARCH

Web-Man – a new software for web-based nutrient management in organic farming

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Abstract An essential prerequisite for the successful expansion of organic farming is an optimal nutrient supply to increase and stabilise yields and reduce the yield gap compared to conventional farming. To date, there is no practical software designed for organic farming, that supports farmers and advisors in soil organic carbon and nutrient management and maps all relevant nutrient flows at the farm level. The development of the nutrient management system Web-Man is intended to close this gap. The paper describes the development, architecture and functions of Web-Man. The system is designed for practical applications in agricultural enterprises and farm advisory services. Web-Man is being developed in a transdisciplinary approach involving software developers, agricultural scientists and future users. The participation of stakeholders enabled the detailed recording of the software requirements of users, researchers and nutrient specialists in government agencies. The Web-Man management system has a modular architecture. The core system comprises the user interface, interfaces for data import and export, and the farm model. The farm model records the structures and production processes of crop and livestock farming. Building on the data collection in the core system, specialized modules for organic farming can be utilized for organic fertilization requirement, humus balancing, on farm nutrient cycling, nitrogen turnover and nitrate leaching and decision support system for crop rotation management. The modules and algorithms take into account important aspects of organic farming, such as long-term fertilizer effects, organic carbon dynamics in soils, and nitrogen transfer in crop rotations. They are tailored to the conditions of organic farming. Web-Man is still under development. The core system and the subject-specific modules 3 Fertilizer requirement calculation (modules 3.1. and 3.2), 4 Nutrient balances (module 4.1), 5 Humus and carbon balances (modules 5.1, 5.2 and 5.3) are already available for end users. Further modules are to be completed by the end of 2025.

Keywords Nutrient management · Digital farming · Organic farming · Nutrient cycle · SOC balancing · Decision support systems

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Introduction

Societal and scientific relevance of the topic

The importance of organic farming in Europe is increasing due to the rising proportion of agricultural land under organic farming (European Commission 2021). By 2030, the share of agricultural land under organic farming is intended to increase to 30% in Germany (Federal Ministry of Food and Agriculture 2019) and 25% in the EU, as organic farming promotes environmentally friendly and resilient production methods, circular economy, and animal welfare (European Commission 2021; Farm to Fork Strategy 2020). An essential prerequisite for the rapid and successful expansion of organic farming is an optimal supply of nutrients to increase and stabilise yields and reduce the yield gap to conventional systems (Muller et al. 2017; Seufert 2019).

Organic farming is subject to restrictions on the use of fertilizers, including the exclusion of mineral-synthetic nitrogen fertilizers and easily soluble phosphorus fertilizers (European Union 2020). Organic farming follows basic principles of nutrient management:

- development and maintenance of high soil fertility and microbial activity as the basis of yield formation (Barnwal et al. 2021; Pathan et al. 2020),
- optimal soil organic matter (SOM) supply (Datta et al. 2020; Pereg and McMillan 2020; Saffeullah et al. 2021),
- indirect plant nutrition through organically bound nutrients (e.g. in compost, farmyard manure) and soil ecosystem processes (e.g. N mineralization) (Frissel 1978; van Mansvelt and Temirbekova 2017),
- largely closed nutrient cycles at farm, inter-farm and regional level (Frissel 1978; Hamm et al. 2017),
- symbiotic N_2 fixation through the integration of grain legumes, legume forage, and legume cover crops in crop rotations and N transfer to non-legumes (Kolbe 2022b; Köpke 1995; Nolte and Werner 1994; van Mansvelt and Temirbekova 2017).

Due to these restrictions and principles of organic farming, significantly lower nutrient surpluses and environmentally relevant nutrient losses occur compared to conventional systems (Chmelíková et al. 2021). With sub-optimal nutrient management, there is a high risk of N losses also in organic farming. For example, high N leaching can occur after a clover-grass ploughing in autumn when there is no significant N uptake by plants (Biernat et al. 2020). On the other hand, negative nutrient balances, leading to a decrease in soil nutrient stocks and nutrient deficiencies, also occur frequently (Cicek et al. 2020; Nolte and Werner 1994; Seufert 2019). According to Cooper et al. (2018), phosphorus deficiency is yield-limiting on 40% of organic farmland in Europe, nitrogen also frequently limits the yield in organic farming (Olesen et al. 2007; Pathan et al. 2020). To ensure adequate nutrient supply to plant stands under these conditions, nutrient cycles, crop rotations, and fertilization must be optimally coordinated – a challenging management task that could be supported by digital management systems. Optimized fertilizer management could increase yields and improve economic efficiency while reducing negative impacts on the climate and environment through lower N emissions (Chmelíková et al. 2021; Hülsbergen et al. 2023).

Digital nutrient management—current status and research needs

Digital management systems are increasingly being used in organic farming and can support farmers and advisors in on-farm soil organic carbon (SOC) and nutrient management. Numerous digital tools and software have been developed for specific applications in organic farming, such as digital field mapping with integrated fertilizer requirement determination (Helm 2024; Schlarb 2021; Stiens 2024), decision support systems for optimized manure application (Brown et al. 2005) or for improved N use efficiency (van der Burgt et al. 2006), models for reducing nutrient losses and greenhouse gas emissions (Feliciano et al. 2017; Mosnier et al. 2017), or simulation models tailored to specific farm types, for example, beef cattle production systems (Machado et al. 2010), organic vegetable production systems (Schad et al. 2023; Fink and Scharpf 1998) or cow-calf production systems (Foley et al. 2011).

One of the most important digital tools in nutrient management is the determination of fertilizer requirement. In Germany, farmers are legally obliged to determine the N fertilizer requirement specific to each field, crop type, and yield according to an algorithm specified in the Fertiliser Application Ordinance (BMEL 2021). However, this algorithm does not take sufficient account of key aspects of organic farming, such as the long-term nitrogen effect of organic fertilizers and the nutrient transfer in crop rotations, e.g., after clover-grass termination.

Various approaches have been developed for nutrient balancing and the analysis and optimization of on-farm nutrient cycles – from very simple farm-gate balances (Halberg et al. 2005) to scientific models that map all relevant nutrient flows in the soil–plant-animal-environment system (Küstermann et al. 2010). The latter have not yet become established in agricultural practice due to their complex structure, high data input requirements, and analysis effort.

Soil nutrient supply and dynamics are closely linked to SOC supply and dynamics. Organic fertilizers not only supply macro- and micronutrients, but also organic matter that contribute to SOC formation and soil carbon sequestration, positively affecting soil properties and functions (Kolbe 2022b; Skrypchuk et al. 2020). Every farmer should therefore be informed about the SOC status and SOM supply of the soils. Various models of different complexity exist for calculating humus[1] balances (Hülsbergen 2003; Kolbe 2010; Küstermann et al. 2010; VDLUFA 2014), but these are not integrated into existing nutrient management tools.

Weckesser et al. (2021) have presented a concept for an on-farm N management system that combines modules (fertilizer requirement determination, N cycle, soil N turnover, decision support system for N transfer in crop rotation) accessing a common database. However, this concept focuses on N, excluding other nutrients and SOC, and it is a conceptual work, not yet been implemented into a software.

To date, there is no practical software specifically tailored to the conditions in organic arable and grassland farming that support farmers and advisors in SOC and nutrient management and maps all relevant nutrient flows in arable crop production, livestock farming, and at the farm level. This gap is to be closed with the development of the web-based nutrient management system Web-Man for organic arable and grassland farming.

Subject and objective of the work

In order to support farmers and advisors in on-farm nutrient management in organic farming, the web-based Nutrient Management System Web-Man has been developed in a transdisciplinary project[2] since 2019. The system is on the verge of practical implementation as part of the system webBESyD,[3] with the first model components scheduled to be deployed in several German federal states from 2025 onwards.

Below, the concept and modular architecture of Web-Man are described:

- the objectives of software development,
- the application areas of Web-Man in practice, advisory services and research,
- the general requirements as well as the special requirements in organic farming for the software,
- the architecture of the system, consisting of a farm model and the subject-specific modules (a) fertilizer requirement determination, (b) nutrient cycle, (c) humus balance and soil carbon sequestration, (d) nitrogen turnover in the soil, (e) decision support system (using N transfer in crop rotation as an example),
- the software implementation.

For each subject-specific module, an overview is provided of:

- the areas of application and objectives,
- the underlying methodology,
- the data requirements and sources, and
- the expected results and outputs of the module.

This paper focuses primarily on presenting the overall Web-Man system. The subject-specific modules are discussed only conceptually at this stage. Modules that are not based on previously validated models will be described in detail in separate publications.

[1] In agricultural practice in Germany, the term humus is used instead of soil organic matter (SOM), and accordingly, humus balance is used instead of soil organic matter balance (Brock et al. 2013).

[2] The project is carried out by the Technical University of Munich, the Weihenstephan-Triesdorf University of Applied Sciences and the Saxon State Office for Environment, Agriculture and Geology, funded by the Federal Office for Agriculture and Food.

[3] webBESyD, funded by the Saxon Ministry for Energy, Climate Protection, Environment and Agriculture and described by Donauer et al. (2025) will be the official software for nutrient management provided to farmers by competent state institutions in Saxony, Saxony-Anhalt, Brandenburg and Thuringia. Since webBESyD and Web-Man use the same IT platform, the Web-Man algorithms for organic farming will be also provided to farmers there as part of webBESyD.

Methodological foundations and objectives of the software development

Development goal, target groups and requirements analysis

The research and development goal is to develop a practically applicable software for use by farm managers and farm advisors that supports optimized SOC and nutrient management in organic farms. The software is designed and developed in a transdisciplinary, participatory approach for application in agricultural enterprises and farm consulting. The use of the software for control purposes (by regulatory authorities) is not intended.

Web-Man is based on the software webBESyD, which was developed for nutrient management in conventional farming (Donauer et al. 2025). WebBESyD contains a core system, which includes a farm model to represent the farm nutrient management, including farm structure and management data. Furthermore, it contains subject-specific modules that comply legal regulations (in Germany) for fertilization as well as advanced, sophisticated modules that support farm nutrient management. Nevertheless, the advanced modules of webBESyD are strongly adapted to conventional farming systems and are not suitable for application in organic farms. Web-Man utilizes these already implemented components of webBESyD, which significantly reduces the development effort and ensures that all legal fertilization requirements implemented in webBESyD are also available to Web-Man. On this basis, Web-Man offers further modules, specifically developed for organic farming.

Requirements elicitation for the software system Web-Man

Capturing the specific user requirements for the software is a crucial part of the development process. This was done in several workshops with stakeholders, which began at the start of the project and continued throughout the project. There are general requirements that affect the successful use of the software in agricultural practice. However, technical requirements specific to organic farming have also been identified (Table 1, (Weckesser et al. 2021)).

Scientific fundamentals and methods

The scientific basis of the system comprises existing and tested methods and models: (a) the BESyD system for determination the fertilizer requirement in organic farming (Kolbe and Köhler 2008), (b) its successor, the webBESyD model for farm nutrient management and counselling in conventional farming (Donauer et al. 2025; BMEL 2021), (c) the REPRO model for analysis of nutrient cycles (Küstermann et al. 2010), and (d) humus balancing methods (Hülsbergen 2003; Kolbe 2010; Küstermann et al. 2010; VDLUFA 2014).

The system concept was based on the principles, algorithms and parameters of the above-mentioned models, which were further developed and brought up to the current state of science and technology. To successfully integrate the different methods into one software, the model parameters were analysed, synchronised and transferred into one system (Module 2, Master data).

Two modules were completely newly developed:

- the module for N fertilizer requirement determination in organic farming (Stieber et al. 2023), and
- the decision support system module for the legume-grass termination and N transfer in crop rotations (Weckesser et al. 2023).

Web-Man implements the specific requirements and restrictions of organic farming and the legal regulations.

The scientific methods and algorithms for determining fertiliser requirements, humus balancing, modelling N turnover and the decision support system for legume-grass termination were derived, tested, and validated in field experiments under different soil and climatic conditions, including:

- field trials on crop rotation and fertilization in organic farming (Kolbe and Meyer 2021; Kolbe 2022a, b),
- long-term field experiments on humus balancing (derivation of model parameters and algorithms, model validation) (Brock et al. 2013, 2012a; Kasper et al. 2015; Kolbe 2010; Leithold et al. 2015),

Table 1 Requirements for a web-based nutrient management system in organic farming (according to Weckesser et al. 2021, supplemented and updated)

General Requirements	Explanations, Implementation in the Software
User Friendliness	– Modern, self-explanatory user interface – Low application effort, one-time entry of farm data – Availability of input data platform-independent usability on all end devices
Transparency	– Transparency of algorithms and model parameters – Centralised data storage – High trustworthiness of the software
Interoperability	– Data import through data interfaces – Simple data export of module results
Data Security and Sovereignty	– Data protection and security – Non-cloud-based web application – Data sovereignty with the farmer
Possibility to develop the Software	– Technical expandability for new components and modules – Development option for continuous adaptation to technical changes
Specific Requirements for Organic Farming	
Consideration of the Nutrient Supply of Organic Fertilizers	– Modelling the long-term effects of organic fertilizers (nutrient supply in crop rotations). Estimation of the N mineralisation of organic fertilizers and deduction from the fertilizer requirement
System Evaluation – Modelling of Nutrient Cycles	– Modelling nutrient flows at the farm system level as an essential part of nutrient management. Analysing nutrient use efficiency and nutrient losses at different system levels
Analysis of SOM Supply and SOC Sequestration	– Humus balancing and modelling of management-dependent changes in SOC stocks
Analysis of N Transfer in the Crop Rotation	– Decision support system for legume-grass based crop rotations – Development of a module for soil N turnover
Compliance with Legal Requirements	– Compliance with all legal requirements from the Fertilizer Application Ordinance, Ordinance on nutrient flow-balances, European union ordinance for organic farming
High dependency of the fertilization requirement, SOC and N turnover on soil conditions	– Consideration of soil conditions (soil type, nutrient content) when determining fertilizer requirements, humus balance and simulating N turnover

- field experiments on the legume-grass termination and N transfer in crop rotations (Weckesser et al., Manuscript in preparation).

A selection of field trials is listed in Table 2.

Transdisciplinary development approach and functional specification

Web-Man is a transdisciplinary project. Each development step is based on agricultural concepts and scientific methods, which have to be precisely implemented in software development. A common modelling language, that can be used by both application experts and software engineers is therefore essential for successful development work, especially for the specification of algorithms. In the field of software engineering, many different approaches to modelling languages, like the unified modelling language (UML), have been developed to model different aspects of a software system. In this project, the critical point is the specification of the agricultural science algorithms. These can be represented very well functionally, namely as an input–output function with an associated process specification. The common modelling language was established through:

- program flow charts, which outline all steps of an algorithm, from data input and determination of master data to the calculation steps,

Table 2 Overview of field trials to derive and validate algorithms and model parameters (selection)

Methods and model parameters of corresponding module	Field trials	Site conditions			Literature
		Location	Soil conditions Soil type Soil-climate region	Climate (1991–2020) mean annual temperature mean annual precipitation	
2 Master data					
Derivation of nutrient contents (N, P, K) in Biomass	Performance of different agricultural systems	Bernburg, 40 km south of Magdeburg	Silty loam loess soils in the arable plain (east)	9,1 °C 469 mm	Brock et al. (2011)
Derivation of nutrient contents (N, P, K) in organic fertiliser and in biomass	Optimisation of extensive agricultural cultivation systems with different livestock numbers	Methau, 30 km north of Chemnitz and Spröda, north-east of Delitzsch, North Saxony	Sandy loam loess soils in the arable plain (east) and loess soils in the transitional areas (east)	8,4 °C 693 mm and 8,8 °C 547 mm	Beckmann et al. (1999) Beckmann et al. (2002) Kolbe (2007) Kolbe (2008)
Derivation of nutrient contents (N, P, K) in organic fertiliser and in biomass	Large plot trial to test a livestock-rich and a livestock-free crop rotation on a loess site	Roda, 40 km northwest of Dresden	Loam loess soils in the arable plain (east)	8,6 °C 711 mm	
Derivation of nutrient contents (N, P, K) in organic fertiliser and in biomass	Comparison of the performance of different crop rotation systems in organic farming	Viehhausen near Freising, 30 km north of Munich and Puch, 30 km north of Munich	Haplic Luvisol silty loam 490 m aSL Bavarian Tertiary Hills	8,0 °C 800 mm	Pommer et al. (2009) Castell et al. (2016)
3 Fertilization requirement calculation					
Validation of methods for determining N requirements	Validation of methods for determining N requirements for winter wheat	Christgrün, 60 km southwest of Chemnitz and Pommritz, 70 km west of Dresden	Clayey loam weathered soils in the transitional areas (east) and loamy sand loess soils in the transitional areas (east)	9,1 °C 572 mm and 10,0 °C 618 mm	Grunert (2020, 2024)
Validation of methods for determining N requirements	Validation of methods for determining N requirements for winter rye	Baruth, 70 km south of Berlin	Sandy Clay loess soils in the transitional areas (east)	10,0 °C 618 mm	Grunert (2020)
Validation of methods for determining N requirements	Validation of methods for determining N requirements for winter barley	Pommritz 70 km west of Dresden and Forchheim 30 km north of Nuremberg	Sandy loam loess soils in the transitional areas and loamy sand Ore Mountains	10,0 °C 618 mm and 8,0 °C 825 mm	Grunert (2020, 2024)

Table 2 (continued)

Methods and model parameters of corresponding module	Field trials	Site conditions: Location	Site conditions: Soil conditions Soil type Soil-climate region	Site conditions: Climate (1991–2020) mean annual temperature mean annual precipitation	Literature
Derivation of long term N supply from organic fertilisation	Site-specific effects of long-term organic farming on arable and crop cultivation and environmentally relevant parameters	Gülzow, 30 southeast of Hamburg	Sandy and sandy clay medium diluvial soils MV and Uckermark	8,2 °C 542 mm	Gruber und Thamm (2008) Gruber (2009)
4 Nutrient balances and nutrient cycles					
Modeling nutrient-cycles, model validation	Organic and conventional farming system Scheyern Analysis of the long-term effects of different farming systems on soils, plants and the environment	Experimental farm Scheyern near Pfaffenhofen, 50 km north of Munich	Various soil types Bavarian Tertiary Hills	8.7 °C 774 mm	Küstermann et al. 2008, 2010) Lin et al. (2016)
N_2 fixation, model validation, model comparison	Field trials with legume crop rotations Comparison of methods for estimating N_2 fixation by legumes and determining the N supply to subsequent crops	Experimental station Thalhausen near Freising, 30 km north of Munich	Haplic Luvisol silty loam 490 m aSL Bavarian Tertiary Hills	8.7 °C 774 mm	Amann et al. (2023, 2025, manuscript in preparation)
5 SOM/SOC balances					
VDLUFA method; Derivation of the model parameters (humus balance coefficients)	20 Long-term field trials	Various locations in Germany	Wide range of soil conditions	Wide range of climatic conditions	VDLUFA (2014)
STAND method; Derivation of the model parameters (humus balance coefficients)	Long term field trials; 471 variants	Various locations in Germany	Group 1: Chernozem; Clay (over 700 mm); Sand (C/N ratios over 12–15) Group 2: Sand – loamy sand (under 8,5 °C); Clay loam, clay Group 3: Sand – loamy sand (over 8,5 °C) Group 4: High sandy loam, sandy loam (under 8,5 °C) Group 5: High sandy loam, sandy loam (over 8,5 °C) Group 6: Loam (C/N ratios under 9,5)		Kolbe (2010)

Table 2 (continued)

Methods and model parameters of corresponding module	Field trials	Site conditions			Literature
		Location	Soil conditions Soil type Soil-climate region	Climate (1991–2020) mean annual temperature mean annual precipitation	
Validation of humus balancing method, simple, advanced and dynamic method, comparison of measured and modelled SOC stocks	Long-term field trial "combination trial" Seehausen Analysis of the long-term effects of manure and mineral fertilizers on soils, plants and the environment	Experimental station Seehausen, 10 km northeast of Leipzig	Stagno-Luvic Gleysol sandy loam 132 n aSL Loess soils in the agricultural plain	9,3 °C 554 mm	Hülsbergen (2003) Donauer et al. (2025)
Validation of humus balancing method, dynamic method, comparison of measured and modelled SOC stocks	Long-term field trial "System Trial" Viehhausen Comparison of organic and conventional farming systems with and without livestock farming	Experimental station Viehhausen near Freising, 30 km north of Munich	Haplic Luvisol silty loam 490 m aSL Bavarian Tertiary Hills	8.7 °C 774 mm	Baumgartner et al. (2022)
Validation of humus balancing method, dynamic method, comparison of measured and modelled SOC stocks	Long-term field trial "Energy Crop Rotations" Viehhausen Crop rotations with different proportions of legumes and maize, organic fertilization with digestates	Experimental station Viehhausen near Freising, 30 km north of Munich	Haplic Luvisol silty loam 490 m aSL Bavarian Tertiary Hills	8.7 °C 774 mm	Simon (2021) Winkhart et al. (2024) Winkhart (2025)
Validation of humus balancing method, dynamic method, comparison of measured and modelled SOC stocks	Long-term trial with compost fertilization Vienna Analysis of the long-term effects of compost fertilization on soils, plants and the environment	Obere Lobau near Vienna	Various soil conditions	Various climate conditions	Erhart et al. (2016)
6 Nitrogen turnover simulation					
Simple method: Derivation of N mineralization	Derivation of an algorithm for the estimation of the biological active time relevant for N mineralization				Franko and Oelschlägel (1995), Franko et al. (1995)

Table 2 (continued)

Methods and model parameters of corresponding module	Field trials	Site conditions			Literature
		Location	Soil conditions Soil type Soil-climate region	Climate (1991–2020) mean annual temperature mean annual precipitation	
Simple method Model validation	Long-term field trial "combination trial" Seehausen Analysis of the long-term effects of manure and mineral fertilizers on soils, plants and the environment	Experimental station Seehausen, 10 km northeast of Leipzig	Stagno-Luvic Gleysol sandy loam 132 m aSL Loess soils in the agricultural plain	9,3 °C 554 mm	Hülsbergen (2003)
Simple method Model validation	On farm field trial with lysimeters	Agricultural farm "Budissa GmbH" 70 km east of Dresden	Stagnic Luvisol, Stagnic Gleysol, Stagnic Gleyic Cambisol, Cambisol Loess soils in the transitional zones (East)	9,9 °C 650 mm	Abraham (2001)
Advanced method: Validation of Seepage water calculation model BOWAM (Dunger, 1985)	Lysimeter station (data from Brandis were also used for model calibration)	Brandis, 20 km east of Leipzig and Buttelstedt, 25 km east of Erfurt and Dürnast near Freising, 35 km north of Munich;	Eutric Cambisol Loess soils in the agricultural plain and Chernozem Loess soils in the transitional zones (East) and Cambisol Bavarian Tertiary Hills	10 °C 608 mm and 9.5 °C 555 mm and 89 °C 818 mm	Donauer et al. (2025)
Advanced method; Derivation of model parameters for organic open-field vegetable production	Field trial with suction cup measurements	Research Station Queckbrunnerhof, 100 km south of Frankfurt am Main	Chernozem 110 m aSL Rhine Plain and Adjacent Valleys	11.3 °C 544 mm + irrigation	Andermann et al. (2013)
7 Decision Support System for the N Transfer of Legume Grass Mixtures					
Validation and improvement of model for N dynamics and crop production system analysis (Seasons: 2021/22, 2023/24, 2024/25)	Legume-grass termination trials; different dates of termination; N uptake of following crop (wheat, maize)	Experimental station Viehhausen near Freising, 30 km north of Munich	Haplic Luvisol silty loam 490 m aSL Bavarian Tertiary Hills	8.7 °C 774 mm	Results will be published at the end of the trial period

- a textual process description that details the process and specific characteristics of the algorithm.

The algorithms are then further refined based on this common language basis. This includes:

- a parameter specification: All parameters used are defined and described in detail in a separate file,
- a test file: One file per defined algorithm with a farm representation and human-calculated results to be used in several software test stages and example farms within the application,
- extended master data: If additional master data is needed for the algorithm, this must be extended,
- a user interface description (mockup): Once the algorithm has been fully implemented and tested, the result display must be shown in the user interface (Figure 3). For this purpose, the desired user interface must be described.

The common modelling language and the refinement of the algorithms based on this common language form an algorithm specification, which is created for each development step.

Software implementation aspects

The focus of the development is on two main areas. The first area includes the development of operational data management and the technical implementation of a core system based on the requirements. For this purpose, the common modelling language is used to conceptualize the operational data management.

The second area of software development focuses on the implementation and use of the algorithms defined on the common modelling language (algorithm specifications). The algorithms are implemented separately from the first subarea described above, adhering to the design principle of separation of concerns. The correctness of each algorithm is ensured by automated software tests. The following software tests are employed:

- Unit tests: Testing individual components of the algorithm,
- Integration tests: Combining the algorithm itself with parts of the first subarea. In integration tests, a farm is created within the system to verify the operational accuracy in combination with the model and farm parameter and the algorithm,
- End-to-end tests: Integration tests in connection with the user interface.
- User tests: Selected test persons test the system for correctness of content, user-friendliness and provide feedback

The test files specified in the algorithm specification are used for testing. To be able to examine the entire software, continuous end-user testing is conducted to test and optimize both areas of software development.

Modern technologies in terms of programming language, software libraries, development tools and the development process are used for software development (Table 3).

Table 3 Software development tools

Programming Language	Java 21, Jakarta 10
Frameworks	Spring Boot, Hibernate, GeoTools, Vaadin
Tools	Git, Gradle, Jenkins, Nexus
Software-Architecture	N-Tier-Architecture (Multitier Architecture)
Database	SQL, Rationale Database
Data Integrations	GIS-Server, Geospatial Vector formats, Identity-Provider, MS Office formats
Application Characteristics	Web-Application, modular, flexible configurable and extendable
Development Process	Iterative, Spiral Model

Architecture and modules of the software

General software architecture

In order to implement the defined requirements (Table 1), structural decisions were made (or adopted from webBESyD) at the beginning of the development of the Web-Man software:

- Web-based software architecture: Development of platform-independent software with centralized data management that can be used from within any well-known web browser. The clients are freed from any update problems, e.g. parameter updates and configuration of external data sources. Maintenance will occur at a single point within the application's backend.
- Modular software architecture (Fig. 1): The foundation of the software system is the core system. The core system contains the master data (model parameters) necessary for the software's operation, the farm model for the digital representation of physical farms, the user interface, and a rights and roles management concept.
- In the farm model, all operational site and management data is collected and stored, the subject-specific modules access this data. This approach enables efficient data collection and ensures consistent calculations (as all subject-specific modules use the same data set). The modular architecture also allows for flexible software expansion by adding additional subject-specific modules.
- Integration of functions of a geographical information system (GIS functions): Use of soil maps and geographical information, possibility for site-specific calculations, evaluations and presentation of results.
- Interfaces to agricultural software and databases: data import and data export to field maps based on a csv interface, import of weather data, import of data (e.g. field identification numbers, field area, field geometry) from the integrated management and control system IACS (Integrated Administration and Control System; Agriculture and rural development 2024).
- Data security and data sovereignty: Based on a rights and role concept, data sovereignty lies with the user (farmer), who can authorize third parties (consultants, soil laboratories) to access the data.

The farm model

The farm model (Model 1, Fig. 1) is part of the core system. In the farm model, all operational data, including general farm information, site and climate data (sub model 1.1), as well as crop production data (sub model 1.2) and livestock data (sub model 1.3), are collected and stored.

Crop production is captured on three system levels:

- Field level: Recording of fields (field name, field contour, size) (example of data entry shown in Fig. 2), either manually or via csv import. Field-specific geographical and soil information such as soil type, SOC content, location in water protection area, soil-climate zone with a geo-data connection. The system is conceptually designed to store and process sub-field-specific data from sensor and satellite data, which can be imported through an interface to calculate sub-field-specific humus and nutrient balances.
- Crop species level: Recording of the cultivated crop species for each field. The crop species for which model parameters are available is selected from the crop species list stored in the master data. Currently, 515 crop species can be selected for organic arable farming.
- Production method level: Recording and documentation of the cultivation method for each crop species and each field. The following field operations are recorded methods:

  - ○ Seeding: Seeding date, seeding rate,
  - • ○ Fertilization: Fertilizer type (currently 178 fertilizers selectable for organic arable farming), fertilizer quantity and nutrient contents,
  - • ○ Harvest: Product type, product quantity, including dry matter content and nutrient contents,
  - ○ Product usage for main and by-products (e.g., straw fertilization, green manure, harvest as fresh matter, fodder preservation) for each crop species and each field.

Livestock farming is recorded at three system levels (Fig. 3):

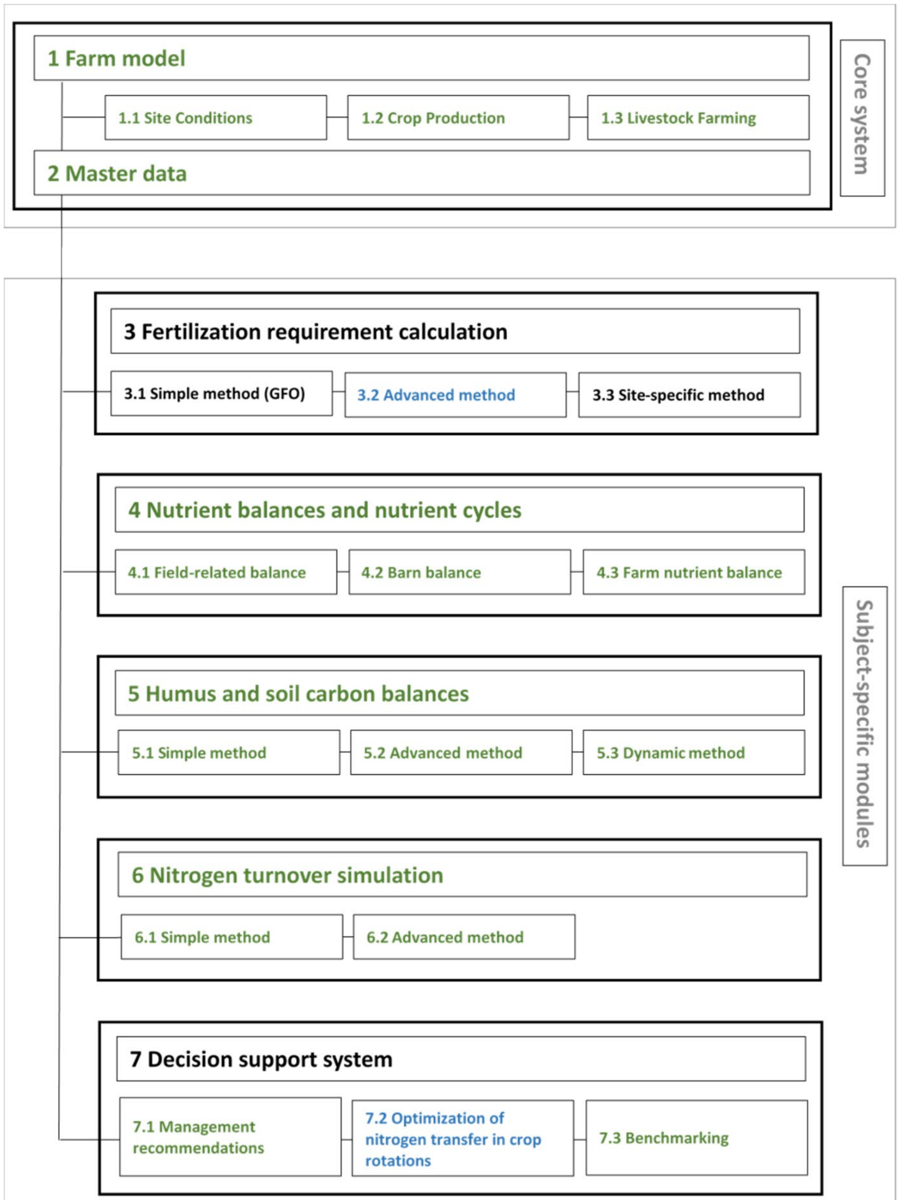

Fig. 1 The modular architecture of Web-Man (Donauer et al. 2025, supplemented): The farm-model manages all farm data, the modules represent the farm nutrient management from fertilizer requirement calculation to estimation of nutrient losses and provide decision support. The different modules access the common data base in the farm model for their calculations. The green parts are adapted to organic farming, the blue parts are newly developed for organic farming

- Barn level: The barn types are categorized according to livestock group and production method (for example, in dairy farming: free-stall with specific stall type, walking area type, pasture access, and milking system) and can be individually adjusted (KTBL 2024),
- Livestock category level: The livestock is categorized, distinguished by livestock group, performance (e.g. milk yield) and potential pasture grazing (Hiller 2014),
- Livestock-related data level (Hiller 2014; Küstermann et al. 2010): All data assigned to a livestock category. This is subdivided into:

  - Herd management: Entries, exits, annual livestock population with changes in herd size.
  - • Feeding management: Detailed information on all feed components used in the feed ration, differentiated between farm-produced feed and purchased feed. The feed components can be

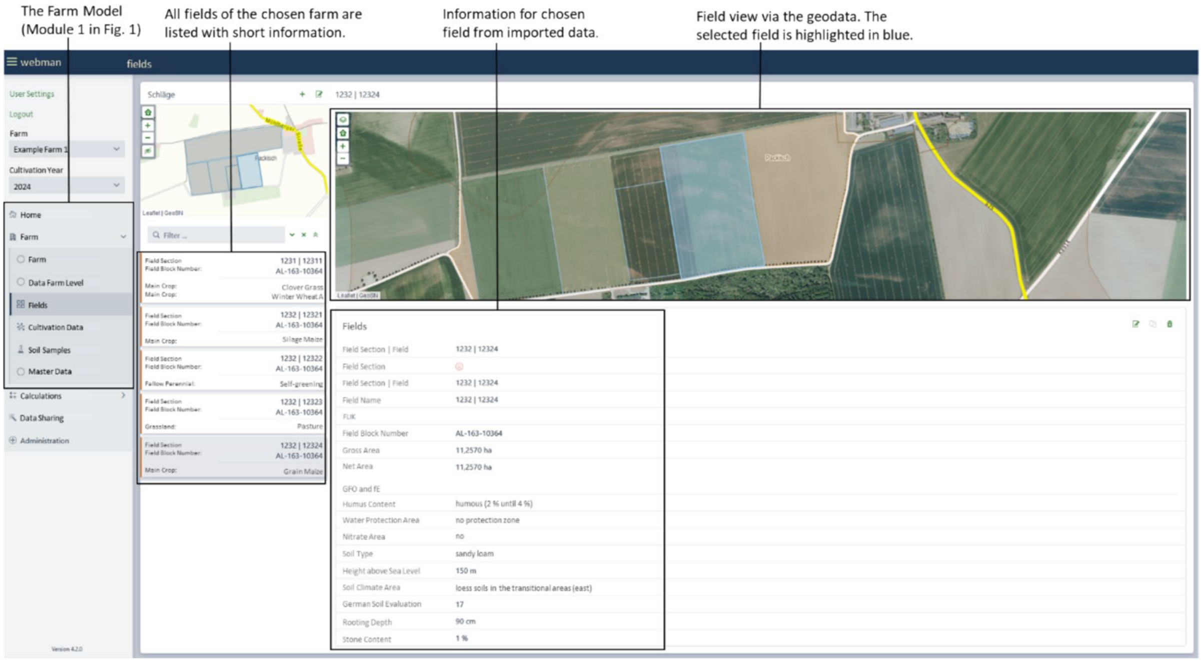

Fig. 2 Data acquisition in the core module (screen display)

selected from a feed list stored in the master data.

- • Livestock performance: Livestock products, changes in live weight.
- Manure management: Assignment of the housing type to the livestock category, type and quantity of bedding material, fertilizer type and fertilizer storage.

Subject-specific modules

Integration of subject-specific modules

The subject-specific modules are based on the core system and the farm model. Each module is built on defined algorithms that retrieve the necessary data (location and management data, master data) from the core system. Technically, each module can be used independently of the others. However, the results from individual modules can also be incorporated into other modules to enhance the information content and precision of the outcomes.

Modules for fertilizer requirement determination

Submodule 3.1, Simple Method According to German fertilizer application ordinance (GFO) (Fig. 1), implements the legal regulations, requirements, and algorithms for determining fertilizer needs. It is also mandatory for organic farms to apply this module to calculate the maximum allowable fertilizer quantity. Field- and crop-specific N requirements are calculated based on target yield, preceding crop, organic fertilization in the preceding year, and the soil mineral N stocks at the beginning of the growing season. The submodule 3.1 is described in detail in Donauer et al. (2025).

For Web-Man, a new (expanded) module for N fertilizer requirement determination for organic farming has been developed (Submodule 3.2), which takes into account the specific conditions of organic farming. A fertilizer requirement calculation for P, K, Mg and lime is currently being developed and is being validated in field trials after completion. The algorithms and parameters are derived based on the simple method according

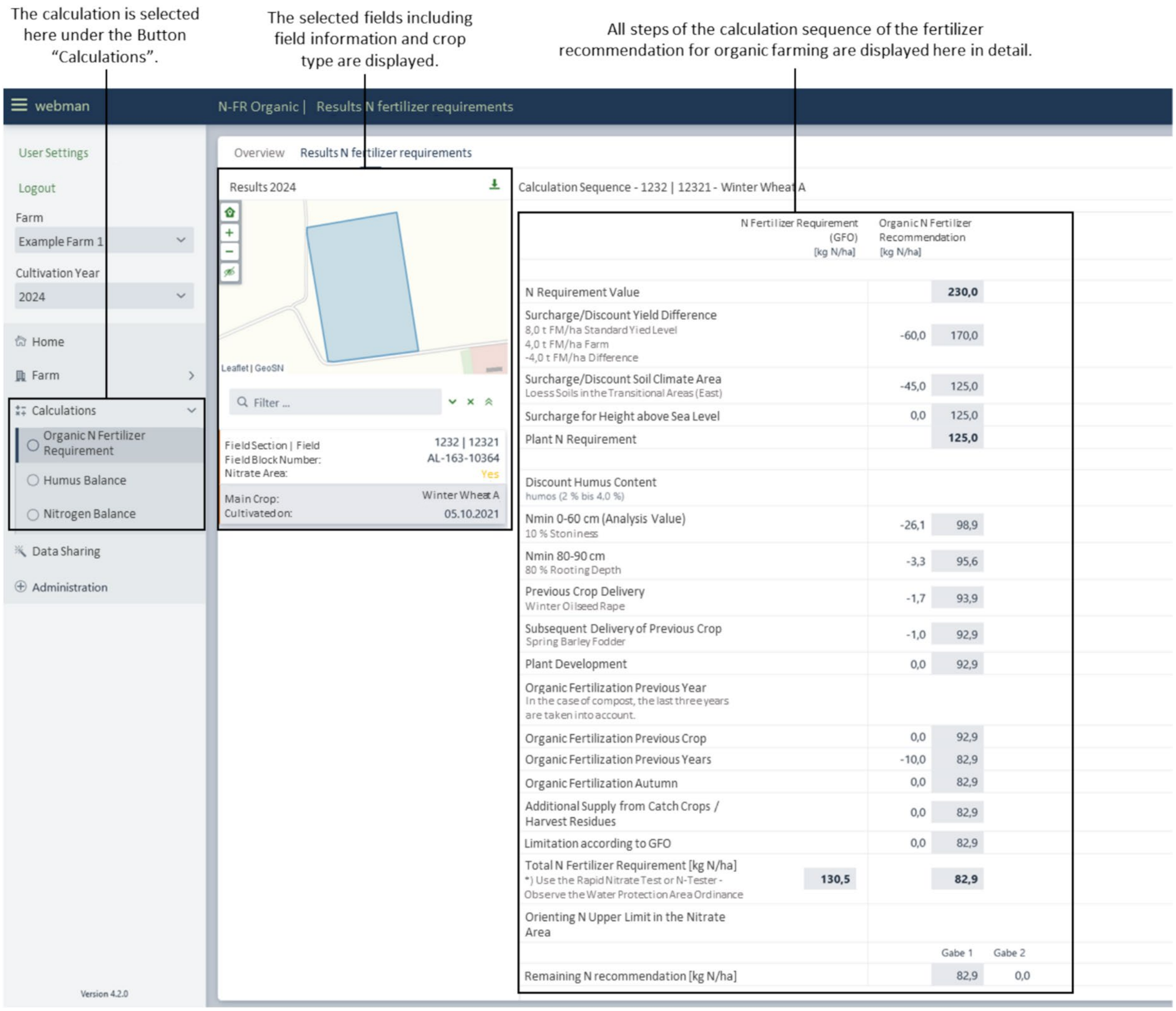

Fig. 3 Result display of the N fertilizer requirement determination (screen display)

to GFO and fertilizer trials conducted under organic farming conditions (Table 2).

The N fertilizer requirement in organic farming is determined using the calculation scheme outlined below (using winter wheat as an example, Eq. 1, with parameter explanations in Table 4):

$$\mathbf{FNR} = NR_{cor} + SCI + Sa + SMN + SD_{pc} + SD_{ppc} + PD + OF_{1\text{-}4} + OF_{a} + CC/GM + WPA + GFO \quad (1)$$

The determination of fertilizer requirements for organic farming takes into account the following:

- N requirement of the crop, summarised in symbol SCI (Table 4): Determined based on target yield and the average N content of organically produced crop products (Kolbe 2022a),
- Climatic and soil-specific influences, summarised in symbol SCI (Table 4),
- N release from the preceding and pre-preceding crops: N release from the preceding and pre-preceding crops depending on crop type, C/N ratio (for preceding crop only), yield, crop duration of the current crop (for preceding crop only), and soil type; up to 25% (10% for pre-

preceding crop) of the N bound in harvest root residues,

- Plant development: Consideration of plant density and BBCH stage (+12 kg N/ha up to −15 kg/ha for low plant density),
- Organic fertilization and its N release potential over the last 4 years: Percentage allocation, in the first year depending on the application month and NH_4-N content on the basis of field trial results from Gruber and Thamm (2008) and Gruber (2009)
- Cover crop cultivation: N release from cover crops and harvest residues (Kolbe 2022b),
- Split applications: If the crop-specific N requirement is exceeded and suitable organic fertilizers for “late” application are available, a recommendation for split applications is made,
- Planned organic fertilization: Indication of already planned fertilization measures, with deductions depending on the type of fertilizer, fertilizer quantity, N_t, and NH_4-N content,
- Verification of legal requirements: If necessary, limitation to the fertilizer requirement according to the GFO; compliance with regulations in water protection areas.

The results of the fertilizer requirement determination according to the GFO and the version for organic farming are displayed to the user in parallel (Fig. 3).

The calculation of P, K, and Mg fertilizer requirements is based on the BESyD/webBESyD methods (Donauer et al. 2025). For the nutrient requirement determination in organic farming, the nutrient contents of harvest products are used, which were derived from field trials in organic farming (Kolbe and Köhler 2008).

Module for humus balancing

The Module 5 Humus and SOC Balances is designed to support farm-level SOC management. It aims to provide users with information on whether the input of organic matter is sufficient to maintain site-specific optimal SOC levels, soil fertility, and productivity, or if agricultural and crop management measures are needed to improve SOC supply. The methods and balance parameters were derived from long-term field experiments.

In Web-Man, three methods of humus balancing are applied: the method according to VDLUFA (2014) (Submodule 5.1), the STAND method according to Kolbe (2010) (Submodule 5.2), and the dynamic humus balancing method according to Küstermann et al. (2010) and Leithold et al. (2015) (Submodule 5.3). The methods differ in terms of data requirements, the factors influencing SOC dynamics that are considered, and the results. However, the basic principle of all three methods is the same—humus balances provide information about management-induced changes in SOC stocks (Hülsbergen 2003). They take into account the effects of crop types (harvest and root residues, rhizodeposition), cultivation methods (soil tillage), and organic fertilization (quantity, quality, and degradability of applied organic fertilizers).

The humus balancing method according to VDLUFA (2014) (Submodule 5.1) represents the simplest approach. The main focus is on quantifying the need for organic matter in crop rotations to maintain soil productivity. A deficiency in organic matter should be avoided in order to prevent negative impacts on soil properties and soil processes and to maintain yield potential. Over-supply of nutrients through organic fertilizers should be avoided to reduce the risk of N losses. For organic farming, specific model parameters were derived from long-term field experiments. Adjusting the model parameters to organic farming is necessary due to the different management practices (e.g. intensive tillage and mechanical weed control, thereby stimulating N mineralisation; lack of mineral N fertilization, higher proportion of soil N in yield formation), which leads to a higher demand for organic matter N input (Leithold et al. 2015).

The VDLUFA method considers

- the effects of crop types on SOC stock in the soil as a combined effect of the input and turnover of organic substances and
- the effects of organic fertilizers on the SOC stock, depending on the quantity, composition, and degradability of the organic fertilizers.

The STAND method according to Kolbe (2010) (Submodule 5.2) is based on the principles of the VDLUFA method but additionally makes a stepwise adjustment to the balance coefficients for organic fertilizers (lower specific effect with increasing fertilizer

Table 4 Definition of the parameters in Eq. 1 for determining N fertilizer requirements with calculation example for high quality backing winter wheat according to Submodule 3.2

Parameter			Example	
Symbol	Name	Explanations	Data	Calculation [kg ha^{-1} N]
FNR	Fertilizer N requirement	N fertiliser requirement, based on plant-available nitrogen		**64,0**
NR_{cor}	N requirement, corrected	Crop and yield-specific N requirement, based on fertiliser application ordinance (BMEL 2021); Backing winter wheat has a fertiliser requirement value of 260 kg ha^{-1} N for a yield of 8 t and 15 kg h^{-1} N are deducted per t)	Backing winter wheat, yield level 5 t ha^{-1}	215,0
SCI	Consideration of soil and climatic conditions	Soil and climatic conditions are taken into account with a surcharge or a discount + adjustment of the N requirement to the target yield and protein content (Kolbe 2022a)	Loess soils in the transitional areas (east), Adaptation of N contents to organic farming	−75,0
SMN	Soil mineral N stock at vegetation start	Soil mineral N stock determined by soil samples and laboratory analyses at the start of vegetation in spring	SMN 0–60 cm: 2% stone content SMN 60–90 cm: 90 cm rooting depth	−24,0 −4,0
SD_{pc}	Subsequent delivery of the previous crop	N supply from crop residues and roots of the previous crop during the cultivation period of the winter wheat	Clover grass, clover content 30%	−18,5
SD_{ppc}	Subsequent delivery of the pre-preceding crop	N supply from crop residues and roots of the pre-preceding crop during the cultivation period of the winter wheat	Clover grass, clover content 30%	−6,4
PD	Plant development	Plant development is taken into account depending on the EC stage and plant density	EC-Stage 22 Plant density: normal	−3,0
$OF_{1\text{-}4}$	Organic fertilization in the last 4 years	The N-mineralization of organic fertilizers of the last 4 years is estimated and deducted from the fertilizer requirement	Organic fertilization from previous years	−10,0
OF_a	Organic fertilization autumn	Organic fertilization already applied in autumn is deducted from the fertilization requirement	No organic fertilization in autumn	0,0
CC/GM	Catch crop/green mass	Additional supply from catch crops and harvest residues	Harvest residues	−10,1
WPA	Consideration of the water protection area	If the field is located in a water protection are: limit the fertilizer requirement to 0 kg ha^{-1} N	No water protection area restrictions	0,0
GFO	Consideration of GFO	The fertilizer recommendation must not exceed the legal regulations	Fertilizer recommendation is below the legal limit (177 kg ha^{-1} N)	0,0

applications). It also takes into account site characteristics for the balance coefficients of crops through corresponding surcharges or deductions.

On the basis of results from long-term field trials, seven site-specific soil groups were defined, which differ in turnover of organic matter in the soil.

The dynamic humus balancing method according to Küstermann et al. (2010) (Submodule 5.3) uses"dynamic"balance coefficients that are adjusted based on site conditions, crop yield, and N uptake by crops (Brock et al. 2012b; Leithold et al. 2015). N balances are used to estimate N mineralization from the soil, and the associated SOC decomposition is calculated (Leithold et al. 2015).

The results of the humus balancing, which indicate potential changes in SOC (Δ SOC)) are evaluated using a rating scheme based on classes, ranging from A (Δ SOC: <-200 kg ha^{-1} yr^{-1}, very poor humus supply) to E (Δ SOC: >500 kg ha^{-1} yr^{-1}, very high SOM supply). The evaluation scheme is adapted to the conditions of organic farming. Based on this evaluation and classification, management recommendations are derived (Module 7.1).

Modules for analysing farm nutrient cycles

The Module 4 Nutrient Balances and Nutrient Cycles is used for analysing farm nutrient flows, including material flows entering and leaving the farm boundaries (Fig. 4). The module also maps denitrification losses according to the method of Hermsmeyer and van der Ploeg (1996) and Köhne and Wendland (1992), nitrate losses according to Küstermann et al. (2010), Renger and Strebel (1980) and Wendling (1993), during crop storage, and in livestock farming, differentiated by their source and process. These nutrient flows link crop production and livestock farming, illustrating interactions (Fig. 4). The module is based on the Repro model (Küstermann et al. 2010) and is only presented as a rough concept here. It is planned that the module will be presented in detail in a separate paper once it has been fully developed and validated.

The analysis and visualization of farm nutrient flows is an effective tool for optimizing nutrient efficiency and minimizing nutrient losses on organic farms. Nutrient cycles can be modelled for N, but also for P, which is particularly important for organic farming due to P deficiency. The analysis and representation of farm nutrient cycles combines various sub-balances (Fig. 4):

- Farm gate balance (Submodule 4.3): This balance records all inputs (purchases of animal and plant products, fertilizers, N deposition, N_2 fixation) and outputs (sales of livestock and plant products, export of organic fertilizers).
- Barn balance (Submodule 4.2): This is used to balance the nutrient flows in livestock farming on the basis of the livestock stock, the housing conditions (KTBL 2024), the milk yield and the feed management. Inputs include farm-produced and purchased feed, straw, and purchased livestock. The outputs include livestock products, and manure. The submodule also estimates gaseous N losses depending on the housing system and management with data from KTBL (2024) and Hiller (2014).
- Field balance (Submodule 4.1): This balances nutrient flows in crop production, recording all field-specific material flows and summarizing them at the farm level. Inputs include mineral and organic fertilizer, N_2 fixation, and N deposition. Outputs are the nutrient removal by crops. Both the organic fertilizers and the nutrient content of the crops are adapted to the conditions of organic farming.

For a transparent and consistent representation of nutrient flows, the stock levels (e.g., feed storage, fertilizer storage) are also recorded, including inputs, outputs, and storage losses, in order to link the material flows from crop production and livestock farming. The Nutrient Cycle Module is highly complex and also integrates results from other modules, such as the dynamic humus balancing adapted to organic farming (Submodule 5.3) and N turnover in the soil (Submodule 6.1), to more accurately quantify N flows (e.g., changes in soil N reserves and nitrate losses).

Module for determination of N turnover and nitrate leaching

Submodules 6.1 and 6.2 calculate N turnover and nitrate leaching from the root zone, taking into account site conditions, crop rotation and fertilization.

Therefore,

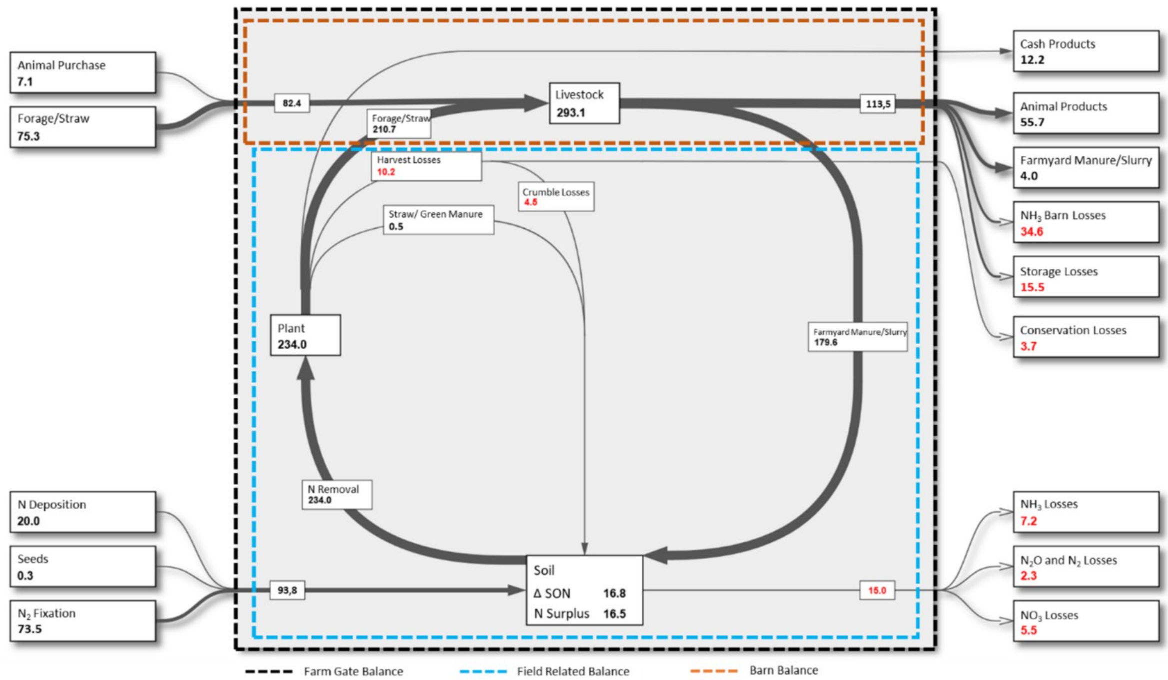

Fig. 4 Nutrient Cycle: The nutrient cycle shows all the material flows used on a farm. It is used to analyse the farm's nutrient situation. The farm gate balance is displayed at the same time as the field-related balance and the barn balance

- the dynamics of plant-available, leachable mineral N in the soil are calculated, influenced by fertilization and mineralization. This involves the consideration of three soil N pools (inert organic N, metabolizable organic-N, mineral-N (Küstermann et al. 2010)), which are interconnected,
- the uptake of plant-available N by crops from the soil is estimated and
- the vertical movement of nitrate N with leachate water is simulated (Submodule 6.2) or calculated (Submodule 6.1).

Necessary soil profile data is available in the German soil map 1:200.000 and its connected database (Federal Institute for Geosciences and Natural Resources 2024; Krug et al. 2013).

Submodule 6.1 utilizes a simple approach, described by Küstermann et al. (2010), in which extended N balances are combined with semi-annual leachate quantities calculated according to Renger and Strebel (1980) and Wendling (1993). This method allows easy calculations for all crop types. It is assumed that the amount of N not removed with crop products, stored in soil organic matter or lost through volatilization as NH_3 or N_2O is at risk of leaching. N inputs soluble N from organic fertilization, N from mineral fertilization, symbiotic N fixation, N deposition and mineralized N from the soil organic N pool. Legume N fixation as important N source in organic farming is estimated crop-specifically based on yield, taking into account the N content in harvested products and crop residues according to the approach of Küstermann et al. (2010).

Submodule 6.2 uses a more sophisticated method, simulating all processes mentioned before on a daily time step. Furthermore, the soil profile is divided into several soil layers (Andermann et al. 2013; Donauer et al. 2025). The BOWAM model (Dunger 2007) is used to calculate seepage. Mineralization of metabolizable organic N is modelled closely following the methodology of Hansen et al. (2012). Key factors are:

- Decomposition coefficients of organic fertilizers, which are differentiated based on whether the N is from rapidly or slowly degradable fractions,

- Clay content of the topsoil, temperature and soil moisture conditions.

In order to estimate the mineralization from the inert N pool, the HU method for humus balancing (Hülsbergen 2003; Küstermann et al. 2010) is used, as in submodule 6.1.

Nitrogen is stored in both above-ground and below-ground biomass. Rooting parameters (rooting depth and root length density) determine from which soil layers N can be absorbed. The root parameters were derived from several measurements made in field trials at our research stations (data not published) using the soil core method (Newman 1966; Schuurman and Goedewaagen 1971; Böhm 1979) and from the literature (Brown and Biscoe 1985; Windt and Maerlaender 1994; Vamerali et al. 2009; Ulas et al. 2012; Svoboda et al. 2020).

Submodule 6.2. requires a detailed parametrisation of each individual crop and each individual fertilizer. Since most fertilizers, such as livestock manure, have different properties in organic farming systems compared to those from conventional farms, separate parametrisation is required. Furthermore, organic farming specific fertilisers need to be parametrised.

Decision support system for the n transfer of legume grass mixtures

A rule-based decision support system provides recommendations for optimizing nutrient management in legume-based cropping systems. The goal is to manage the plant-available N after a legume-grass termination to ensure a low-loss N transfer to the subsequent crop (Weckesser et al. 2024). Depending on the growth, the amount of fixed N, including shoot and root residues, can range from 100 to 400 kg N ha^{-1} (Anglade et al. 2015; Barbieri et al. 2023; Kolbe et al. 2006; Loges et al. 1999). The decision variables include, the management of the legume-grass biomass (removal or green manure), the timing of termination, the subsequent crop and its sowing date, as well as organic fertilization.

The decision support system (Submodule 7.2) first evaluates the nitrate leaching risk at the site. The underlying model considers field-specific mineralization and leaching potential. This includes soil parameters (e.g., soil type, rooting depth, and soil–water balance parameters) and management parameters (e.g., mineralization potential, termination timing, expected yield). The processes

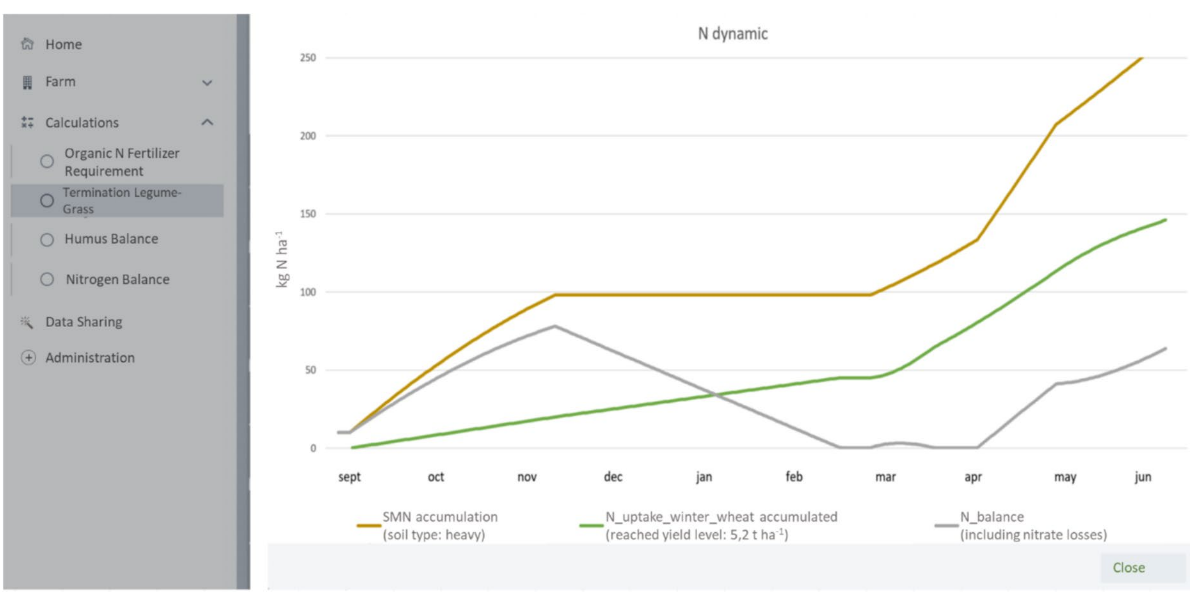

Fig. 5 Soil mineral N (SMN) dynamics of an autumn termination with following sowing of a winter crop. Termination date of legume-grass layer: 15th September. Sowing date of winter wheat: 20.th October. The plot is showing a considerable N loss through nitrate leaching in the winter term and a reduced N uptake in March and April because of lacking N availability. N mineralization progression includes an organic fertilization in April (demonstration model version 04/2024)

(mineralization, leaching, N uptake) are simulated and assessed, taking into account the user's planning data (Fig. 5).

An additional visualization of the SMN pool serves to raise awareness about the termination decision, including the follow-up assessment of possible management measures.

Further explanations of the model:

- Parameters used for the N mineralization potential: Growth-related parameters (composition of the legume-grass stand, crop yield) must be recorded or estimated by the user. Based on the amount of N contained in the legume-grass biomass, a mineralization process is predicted, considering the vegetation dormancy (permanent fall below 5 °C in the first or last week of the year) at the site. As the relevant time series data may be several weeks in the future and weather pattern is unpredictable, the used year-independent function is limited by these uncertain parameters.
- Classification of nitrate risk: A long-term percolation water map (Wessolek and Duijnisveld 2003) and a day-specific soil moisture map (Web Map Service from DWD (2025)) are used. The soil moisture at the planning date is normalized to the long-term soil moisture at the planned termination date, ensuring that annual effects are sufficiently considered. Depending on N mineralization, an assessment of N leaching risk (nitrate concentration, nitrate load) at the site is conducted.
- N uptake of the subsequent crop: The N mineralization from the legume-grass biomass and the applied or planned organic fertilization, along with the potential N uptake of selected crops, are balanced daily. N uptake is mapped using a physical model, with algorithms parameterized according to site conditions (climate, yield expectations) and management practices (sowing date, N availability). For winter crops, by dividing N uptake into different growth stages, compensatory effects related to sowing date and N supply can be represented.

Summarized, the algorithm aims to approximate the N mineralization process, N loss, and the N uptake process of the subsequent crop to provide a recommendation for different optimization goals (maximum economic yield, minimal N loss risk, compromise), that can be chosen by the user. Therefore, for every possible termination and sowing date combinations, the resulting nitrate load, time series of N dynamics, and a yield forecast are allocated. At any time during the module utilisation, users are allowed to change parameters serving to iteratively simulate the effects of management decisions (e.g., postpone the termination date, change the choice of the subsequent crop, reduce biomass before ploughing).

Finally, the model carries out a day-based simulation of N dynamics. All relevant variables, including N mineralization, N uptake, nitrate load, and yield losses (e.g., due to possible postponement of optimal measurement dates at the expense of potential yield and in favour of minimizing the nitrate load), are simulated per day. The subsequent derivation of the optimized termination and sowing date combinations, depending on the optimization target, is also based on these sample spaces. Furthermore, additional textual guidance is given.

Optionally, users can simulate termination scenarios by adjusting management data (e.g., changing the termination date, plan organic fertilization) to support decision-making for farm nutrient management. The Submodule 7.2 is conceptually described here and is currently under development.

Discussion

System approach and interoperability

The goal of the Web-Man project is to develop a practical software tool that integrates various nutrient management functions for organic farming into one system. This is in line with the widespread target in organic farming of system analysis and system optimization (van Mansvelt and Temirbekova 2017). Web-Man allows the integration of data from different farm sectors to represent nutrient cycles and illustrate the interactions between crop production and livestock farming. Web-man is based on established methods for modelling farm nutrient cycles (Hülsbergen 2003; Küstermann et al. 2010). These approaches have been further refined in Web-Man aligning them with the latest advancements in science and technology.

In contrast to previous individual solutions for special applications, users now only need to familiarize themselves with one software system. Data required for nutrient management is collected once in the farm model and then applied to various subject-specific modules. The farm model aims to address the persistent issue of insufficient interoperability in the agricultural sector, at both the technical and operational level (Roussaki et al. 2023). Web-Man can be integrated into other programs through an established interface, while it also utilizes datasets and algorithms from other agricultural software via developed interfaces. The user requirements for interoperability, as outlined by Roussaki et al. (2023) in their data and interoperability concept, have also been implemented as far as possible in Web-Man.

A fundamental feature of Web-Man is that it offers modules with different levels of technical depth and complexity. Simplified methods for fertilizer requirement calculation, nutrient and humus balancing provide an easy entry into the software and support the implementation of legal regulations. More advanced modules, requiring greater technical expertise and data input, enable deeper analysis. Methods for site-specific nutrient management (Hagn et al. 2024; Mittermayer et al. 2022; Pawase et al. 2023), which also have significant potential for application under organic farming conditions (Schuster et al. 2023), have been planned conceptually but not yet implemented.

Quality through extensive software testing

Software testing is one of the most critical components in software development for ensuring quality (Aniche 2022; Spillner and Linz 2021). Compared to desktop applications, testing of web applications like Web-Man is considered complex due to the system's dynamic nature. At the same time, however, the adaptation effort is lower (Di Lucca and Fasolino 2006). Web-Man uses a four-stage test model (unit tests, integration tests, end-to-end tests, and user tests), which, although resource-intensive, is essential for quality assurance (Aniche 2022; Di Lucca and Fasolino 2006). Automated testing procedures help minimize testing effort (Bons et al. 2023).

A particular focus in the development of Web-Man is on user tests. In this test, users are given the opportunity to apply the system in order to obtain feedback from the user's perspective regarding usability, plausibility and the presentation of results, as well as the applicability of management recommendations (Barnum 2021). This feedback is then used to further optimise the software, ensuring the successful implementation and acceptance in practice.

Web-based system

Web-Man was developed using a web-based architecture, which offers significant technical and user-related advantages (Jahanshiri and Shariff 2014). A key benefit of a web-based application is its future viability in terms of data analysis and efficient data management, allowing for seamless integration into user-friendly management systems through API interfaces (Dastres and Soori 2021; Jahanshiri and Shariff 2014). In recent years, various agricultural software developments have used web-based systems, primarily due to their user-friendliness (Dall'Agnol et al. 2020; Dewi et al. 2021; Medici et al. 2021; Shamrat 2020; Malallah and Abdulrazzaq 2023).

Implementation of the specific features of organic farming in the management system

A primary objective of the Web-Man project was to optimally adapt a digital nutrient management system to the conditions of organic farming and to develop specific modules for organic agriculture.

The characteristics and goals of nutrient management in organic farming described at the outset were implemented in Web-Man as follows:

- Limited nutrient availability due to restrictions in organic farming and partially low soil nutrient levels (Cicek et al. 2020; Seufert 2019): Determination of fertilizer requirements (Module 3) for the most efficient use of scarce and expensive nutrients and fertilizers, thus avoiding over-fertilization and increasing fertilizer use efficiency.
- Optimal SOM supply to improve soil fertility and yield capacity (Datta et al. 2020; Saffeullah et al. 2021): Integration of humus balancing methods (Module 5) for analysing and assessing management-induced SOC changes and providing recommendations for on-farm SOC management.
- Indirect plant nutrition through organically bound nutrients and nutrient transfer in crop rotations (van Mansvelt and Temirbekova 2017): Analysis of N turnover (Module 6) and decision support for legume-grass termination to increase N efficiency and reduce N losses (Submodule 7.2).
- Largely closed nutrient cycles on farms and reduction of environmentally relevant emissions (Hamm

et al. 2017): Modelling nutrient cycles within the soil-plant-animal-environment system (Module 4) by combining nutrient balancing in crop production (Submodule 4.1), livestock management (Submodule 4.2), and farm-gate balances (Submodule 4.3).

Challenges, limits and possible future problems

The development of Web-Man was and continues to be associated with major challenges. In order to combine the various validated and tested models in one system, model parameters had to be developed into a specific system that can include the extensive parameter data required for all models. The legal requirements also had to be taken into account so that farmers could use the system. Due to missing or outdated digitalisation, this required a great deal of work.

Furthermore, Web-Man is attempting to bring together some calculations. Further modules are already planned and the system has also been developed to be expandable. Nevertheless, there are certain limits to the expandability of this system and it will not be possible to integrate every model into Web-Man.

In addition, the software must be maintained and serviced, political changes must be incorporated, model parameters must be regularly checked and corrected according to current scientific results and new scientific findings must be incorporated. Only under these circumstances can Web-Man be used in agricultural practice in the long term.

Conclusion and Outlook

The development of Web-Man is still ongoing and will continue in the future, as it is a dynamic, adaptable, and flexibly expandable system. This is partly due to potential changes in fertilization regulations, the continuous advancement of fertilization and modelling systems, and the introduction of new technologies. As the number of users increases, there will be growing feedback for maintaining and further developing the software. Therefore, sufficient capacity for hosting, maintenance, support, and system development must be ensured for the software's long-term operation.

The modular system architecture, in which the core system, including the farm model, is combined with subject-specific modules, offers opportunities to integrate additional functionality as new modules, such as a greenhouse gas balance, which is increasingly being demanded in food value chains. A major advantage is that most of the necessary data is already available in the system.

Web-Man was developed with great care, using the latest scientific knowledge in agriculture and modern software technology. Given the increasing importance of digital systems in agriculture, Web-Man has the potential to become a comprehensive nutrient management system for organic farming.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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